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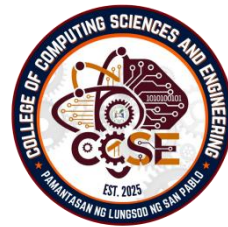
PAMANTASAN NG LUNGSOD NG SAN PABLO
**COLLEGE OF COMPUTING SCIENCES
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ELECTIVE 2

BSCPE-3D

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Figure 1. Hikers Banner

HIKERS is a web-based hiking booking platform that allows users to explore hiking destinations, view schedules, and reserve hiking trips online. It provides a simple system for managing bookings for both users and administrators.

Key Features

User Features

- User Registration and Login
- Explore Hiking Destinations
- View Hiking Schedules
- Make Hiking Reservations
- Add Mountains to Favorites
- View Booking History
- Interactive Maps and Location Viewing
- User Achievements and Hiking Progress
- User Profile Management
- Submit Reviews and Ratings
- Community Feature for Sharing Hiking Experiences
- Safety Tips and Hiking Guidelines

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Admin Features

- Admin Dashboard
- Approve or Reject Bookings
- Add and Manage Hiking Destinations
- Manage User Accounts
- View Booking Records
- Security Features

Authentication System (Login/Register)

Role-Based Access Control (User / Admin)

Form Validation

System Architecture

The HIKERS platform follows a Client-Server Architecture.

Frontend

- Built using React.js, the frontend allows users to:
- Browse hiking destinations
- View schedules
- Make reservations
- Manage bookings

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Backend

- Developed using Node.js and Express.js, the backend handles:
- API requests
- Booking management
- User authentication
- Database operations

Database

- The system stores:
- User accounts
- Hiking destinations
- Booking records
- Reservation details

Deployment

The system was originally set up on an Ubuntu Server OS using a configured web server and FTP/SCP for file transfer and deployment during development.

For easier access and presentation purposes, the frontend of the system is also deployed on Vercel, allowing users to access the application through a public link.

- Frontend (Vercel): <https://hikers-mu.vercel.app>
- Server (Ubuntu VM): Used for backend deployment and development testing

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Technologies Used

Technology	Purpose
React.js	Frontend
Node.js	Backend
Express.js	Server Framework
MongoDB	Database
CSS	Styling
JavaScript	Logic

Project Development Team — Group 5

JANNA CLaura – UI/UX design and frontend development

HAZEL GAPAScan – Frontend support and form implementation

CAMILLE SINAG – Backend and database support

CASEY DAYO – Testing and debugging

NEIL RUSSEL RUBIA – Backend logic assistance

ELOISA SALVAGO – Documentation and project coordination

SENDERICK ESEO – Deployment and system integration support

Future Improvements

- Improved booking calendar system
- User feedback and rating system
- Notification system *“Primed to Lead and Serve for Progress”*



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User Features

User Registration and Login

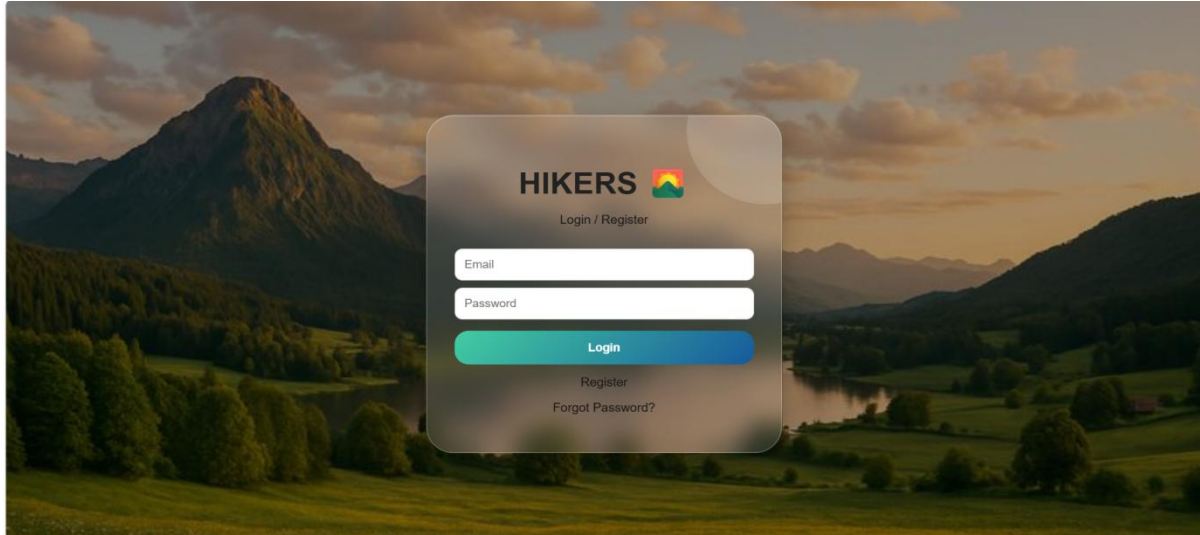


Figure 2. User Registration and Login

Users can create their own accounts and securely log in to access the platform's features. Registered users can manage their hiking activities more efficiently. This feature also allows users to save their preferences and track their progress within the platform.

Welcome Page

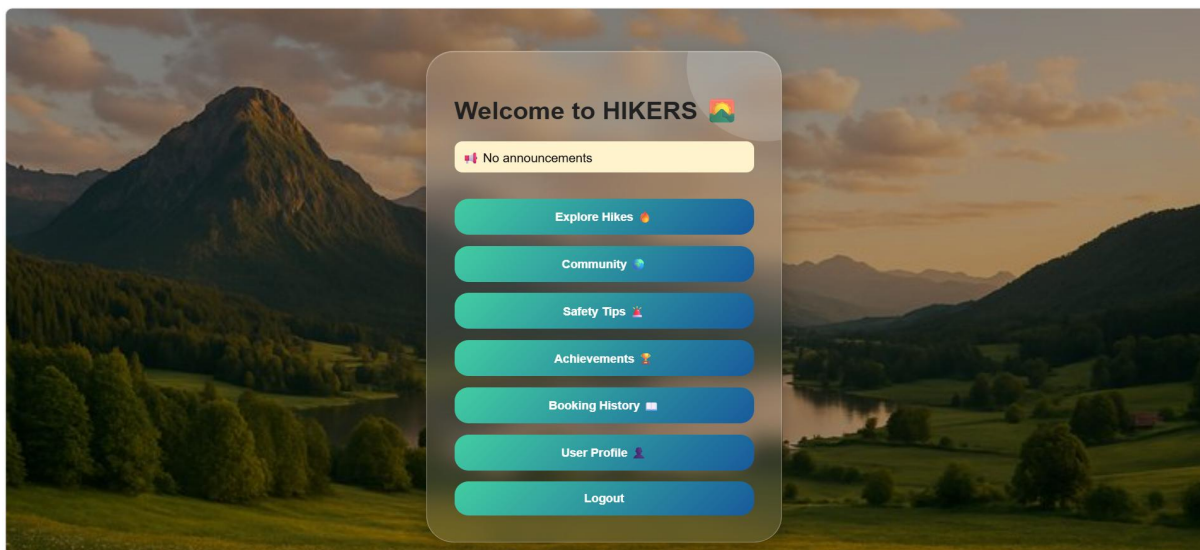


Figure 3. Welcome Page
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The Welcome Page serves as the main landing page of the HIKERS platform where users are introduced to the website and its features. It provides an overview of the hiking booking system, including available services, featured destinations, and important information for users. The page is designed with an attractive and user-friendly interface to encourage visitors to explore the platform. It also provides easy navigation to login, registration, destination browsing, and other major sections of the website.

Explore Hiking Destinations

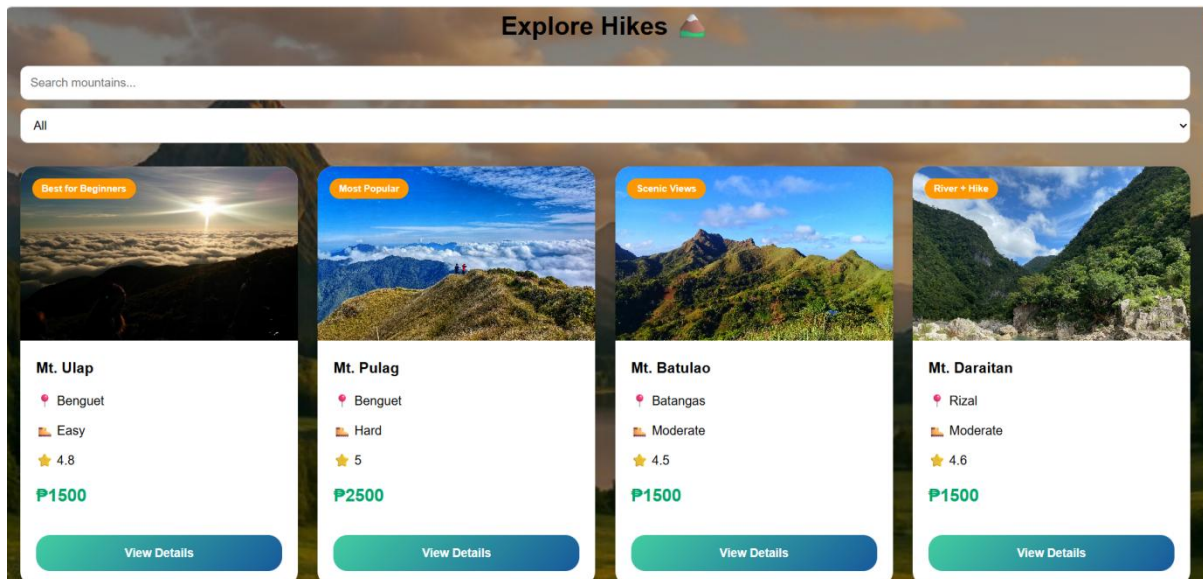


Figure 4. Explore Hikes

The platform allows users to browse various hiking destinations along with important details about each location. Users can view descriptions, difficulty levels, and other information to help them choose suitable hiking trips. This feature makes exploring mountains more organized and accessible. It also encourages users to discover new hiking experiences.

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Mountains Description

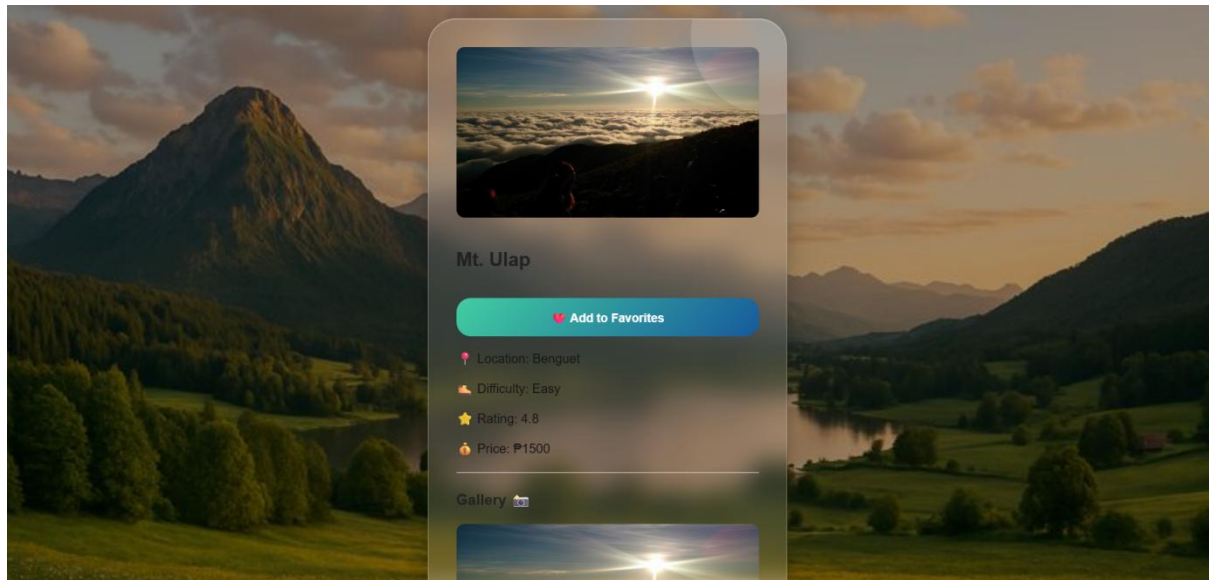


Figure 5. Description

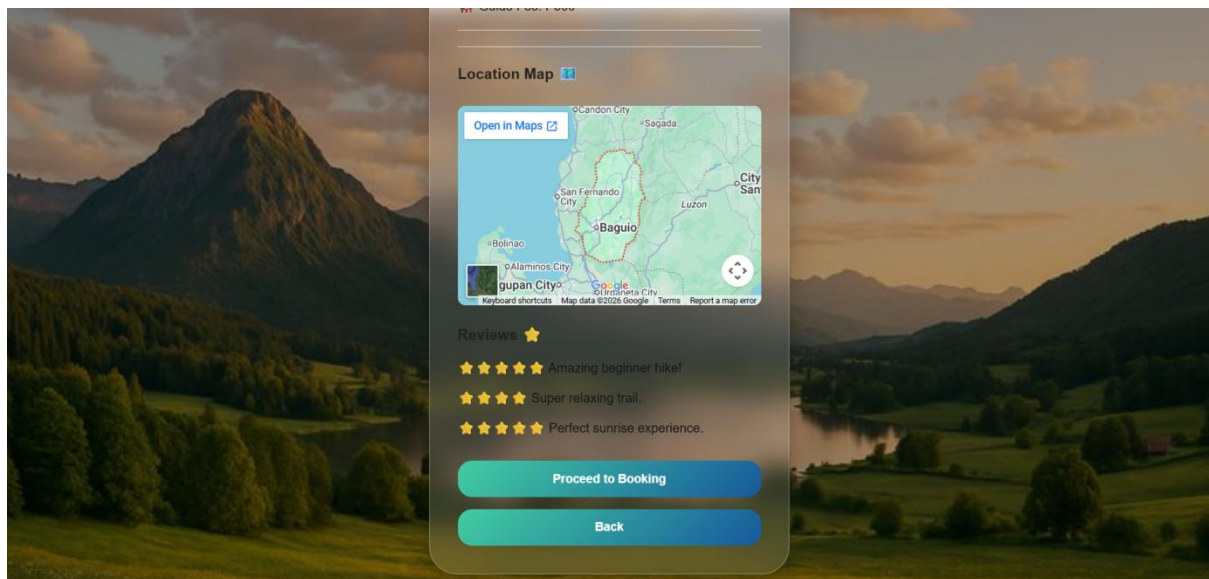


Figure 6. Interactive Maps

- Add Mountains to Favorites

Users can save their preferred hiking destinations using the favorites feature. This allows them to quickly access mountains they are interested in visiting in the future. It improves user convenience by personalizing their experience on the platform.

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The feature also helps users keep track of destinations they want to explore.

- Interactive Maps and Location Viewing

Interactive maps help users view the exact location of hiking destinations. This feature provides better navigation and location awareness before planning a trip. Users can easily understand where the hiking areas are located. It also improves accessibility and trip preparation for hikers.

Schedule Hike and Payment

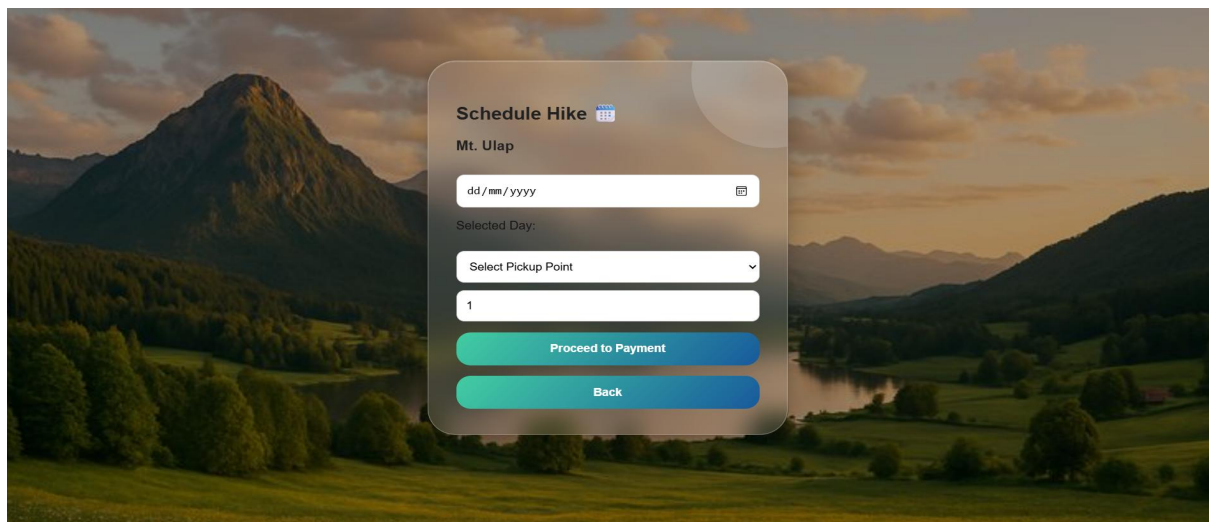


Figure 7. Schedule Hike

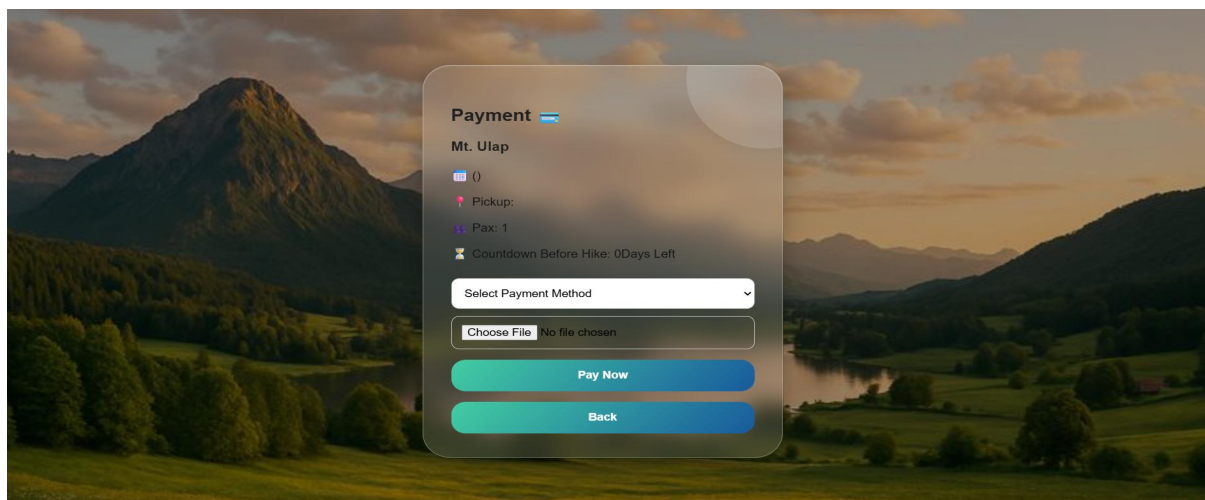


Figure 8. Payment

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The Schedule Hike and Payment feature allows users to select their preferred hiking trip and complete the reservation process through the platform. Users can choose available schedules, review trip details, and proceed with payment for confirmation of their booking. This feature helps provide a more organized and convenient reservation experience for hikers. It also improves transaction management by ensuring that bookings and payments are properly recorded within the system.

View Booking History

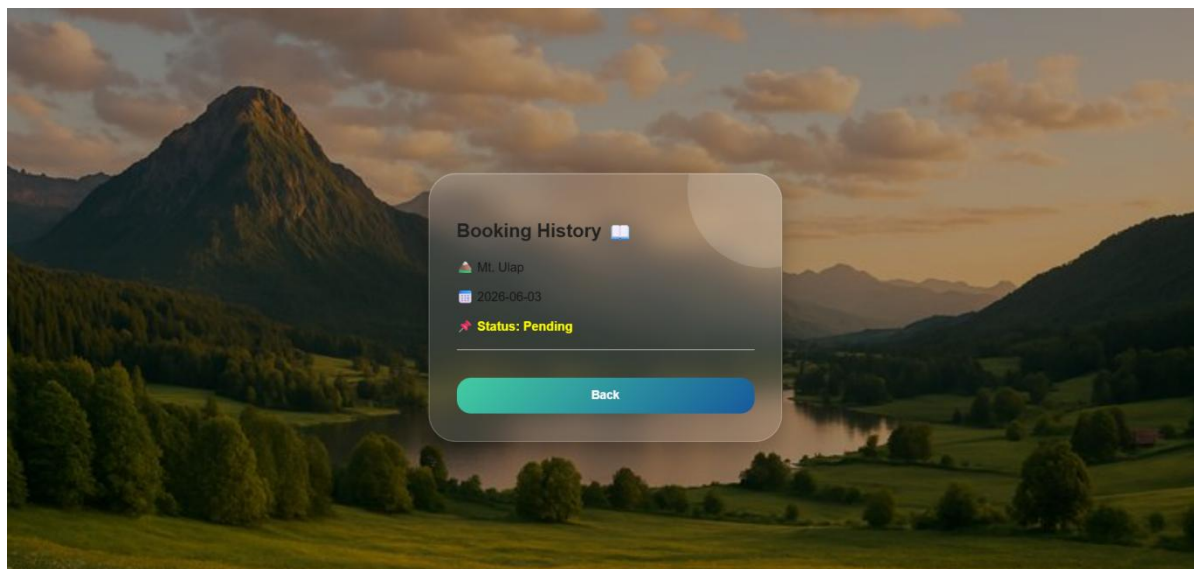


Figure 9. Booking History

The system allows users to review their previous and current hiking reservations. This feature helps users monitor their booking activities and trip participation. It provides easy access to reservation records for reference purposes. Users can also use this feature to track their hiking experiences over time.

User Profile Management

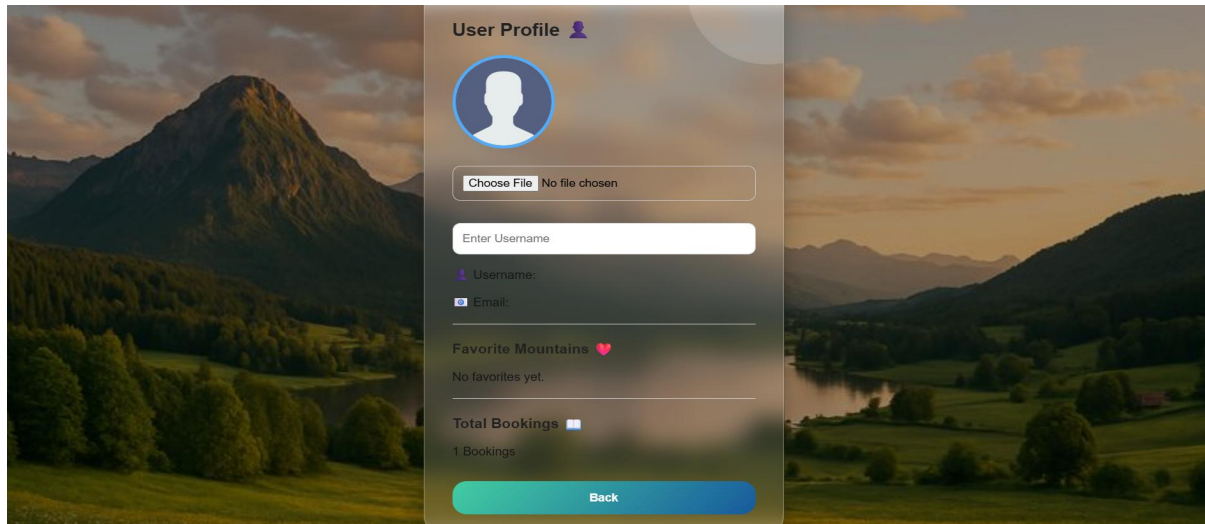


Figure 10. User Profile

Users can update and manage their personal information through their profiles. This feature allows them to customize account details and maintain accurate information. It improves account organization and user convenience. Users can also manage profile-related settings within the system.

User Achievements and Hiking Progress

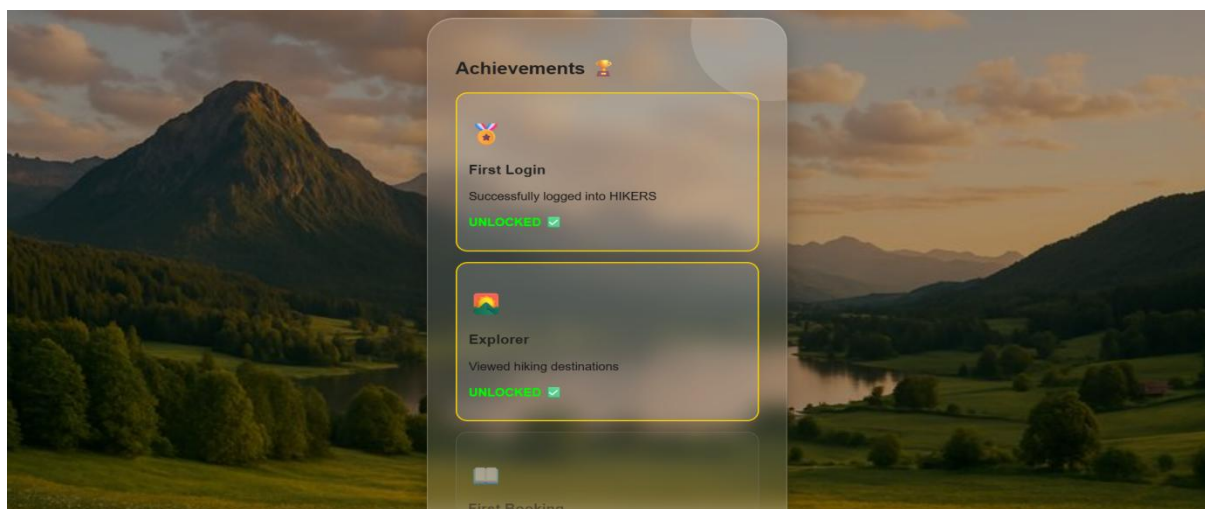


Figure 11. User Achievements

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The platform includes an achievement system that tracks users' hiking activities and milestones. This feature motivates users to participate in more hiking trips and engage with the platform. It provides a more interactive and rewarding user experience. Users can also monitor their hiking journey and accomplishments.

Community Feature for Sharing Hiking Experiences

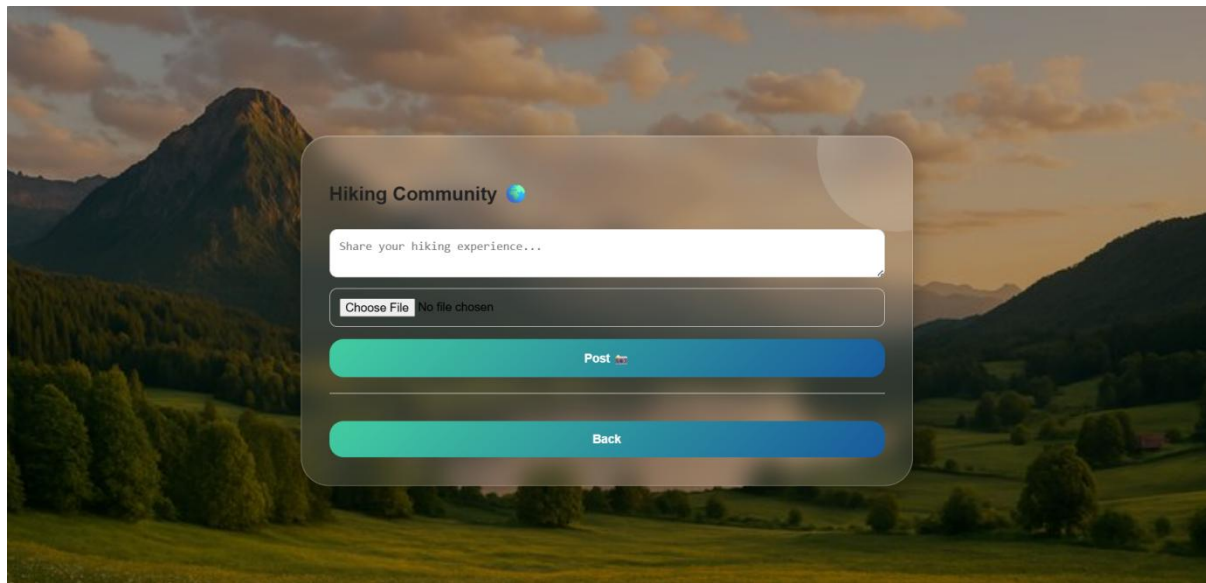


Figure 12. Community Feature

The community feature allows users to share hiking stories, experiences, and updates with other hikers. It promotes interaction and engagement among members of the platform. Users can connect with fellow hikers and exchange useful information. This feature helps build an active and supportive hiking community.



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Safety Tips and Hiking Guidelines

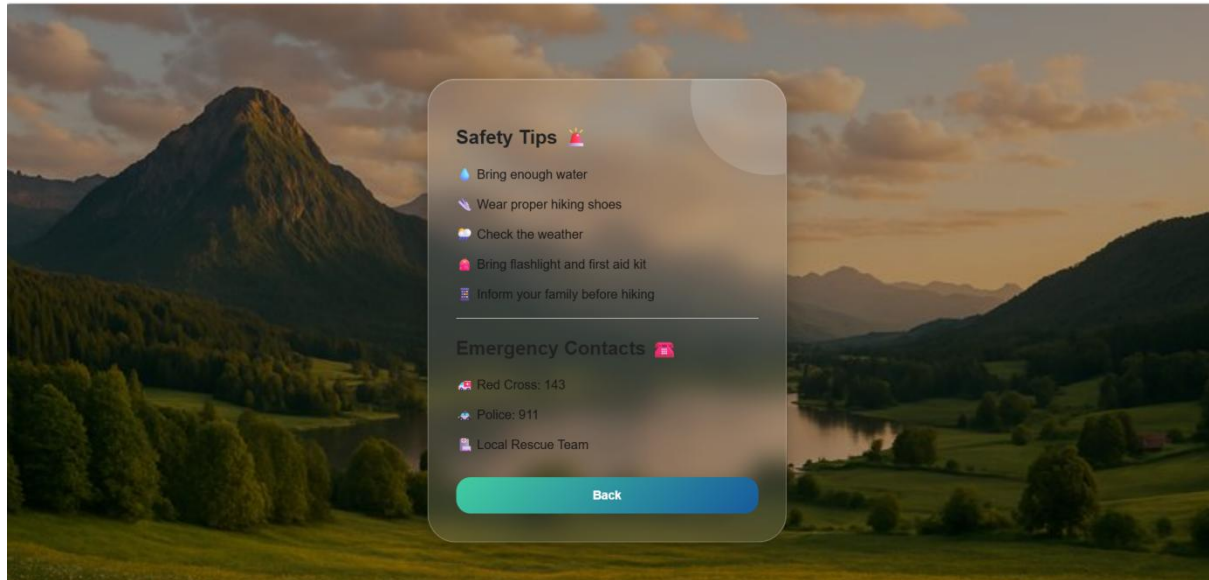


Figure 13. Safety Tips

The platform provides safety reminders and hiking guidelines for users before joining trips. This feature helps promote responsible hiking and proper outdoor preparation. Users can learn important safety practices and precautions. It supports safer and more informed hiking experiences.

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Admin Features

Admin Dashboard

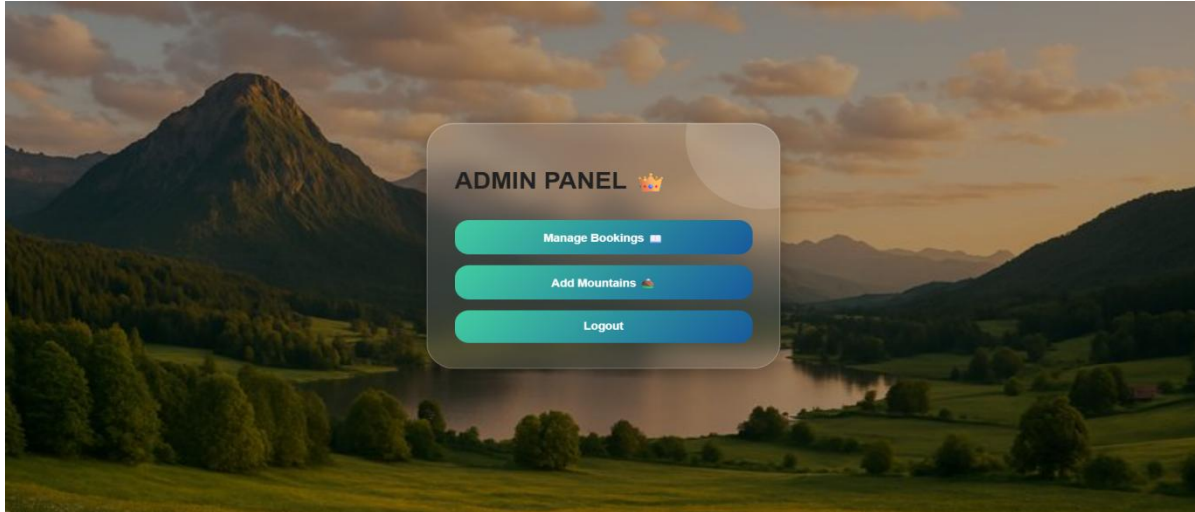


Figure 14. Admin Panel

The admin dashboard provides administrators with an overview of system activities and booking information. It helps admins monitor reservations, users, and destination records efficiently. Important data and management tools are organized in one interface. This feature improves system administration and overall platform management.

Approve or Reject Bookings

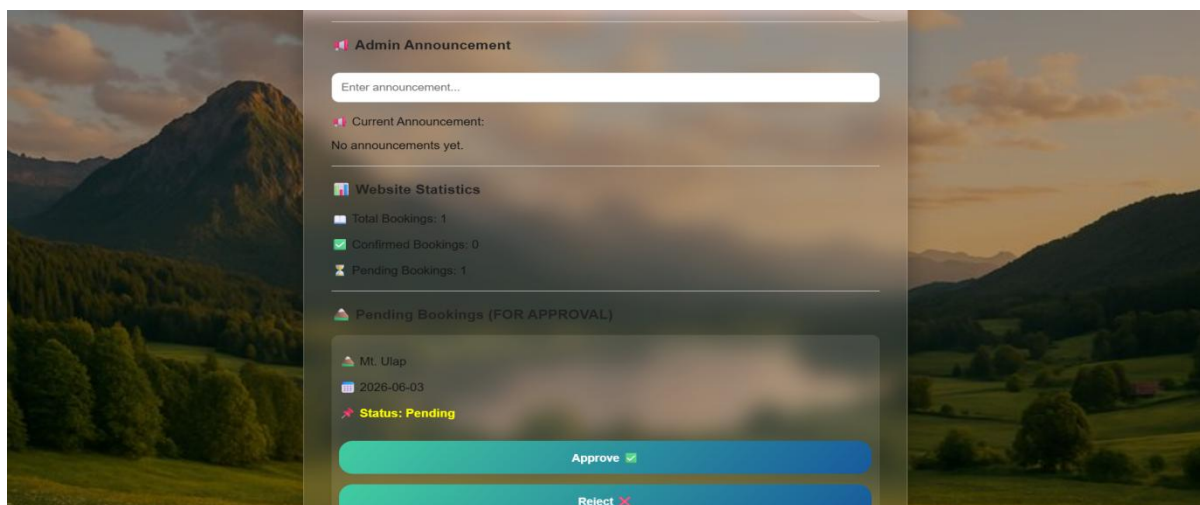


Figure 15. Approve or Reject Bookings

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Add and Manage Hiking Destinations

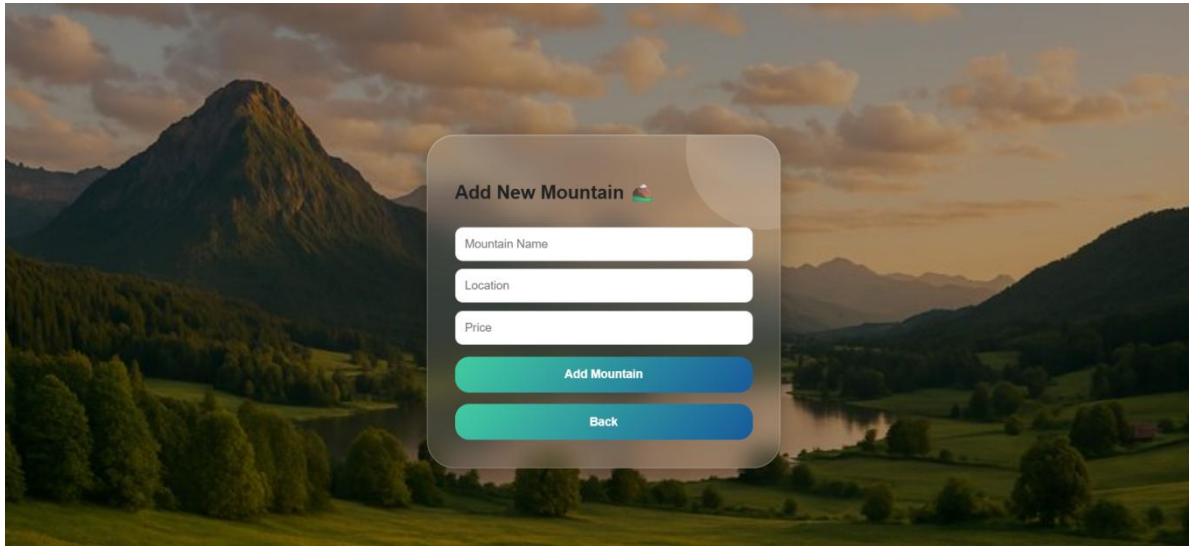


Figure 16. Manage Mountains

Admins can add new hiking destinations and update existing information within the platform. This feature ensures that users receive accurate and updated destination details. Administrators can manage schedules, descriptions, and related hiking information. It helps maintain the quality and reliability of the platform's content.

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